

GUJARAT TECHNOLOGICAL UNIVERSITY
BE –SEMESTER 1&2(NEW SYLLABUS) EXAMINATION- WINTER 2018

Subject Code: 3110004**Date: 10-01-2019****Subject Name: BASIC CIVIL ENGINEERING****Time: 10:30 am to 01:00 pm****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Discuss Branches of Civil Engineering	03
	(b) Enlist principles of planning and describe any one in detail	04
	(c) Enlist different types of foundation and explain Open footing	07
Q.2	(a) Explain Application of Total station	03
	(b) Explain features of smart cities	04
	(c) The following sequential readings were taken on a sloping ground . The first reading was taken on a BM of 50.000 m . Find out RL of each point and apply arithmetical check 0.450,0.560, 0.785,1.200,2.500,3.480,0.320,0.655,0.985,1.300	07

OR

- (c) The following bearings were taken on a closed traverse ABCD find out internal angles and correct them if necessary **07**

Line	FB	BB
AB	55°30'	235°00'
BC	113°00'	293°30'
CD	190°00'	10°00'
DA	279°00'	99°00'

Q.3	(a) Enlist Instruments used in chain surveying	03
	(b) Draw sketch of Dumpy level and show all parts	04
	(c) Give classification of solid waste and describe why its segregation in wet and dry is necessary ?	07

OR

Q.3	(a) Describe primary division of surveying	03
	(b) Draw traffic signals for (i) speed limit (ii) No parking	04
	(c) Which are the water quality standards . Give values of water quality standard for PH and Total dissolved solids as per IS 10500:2012	07
Q.4	(a) Describe house water connection with sketch	03
	(b) Give difference between Surveyors and prismatic compass	04
	(c) Enlist different bonds of Brick and Describe English bond in detail	07

OR

Q.4	(a) Draw sketch of RCC beam	03
	(b) Explain Aim ,objectives and key features of BRTS	04
	(c) Explain Energy efficient Building	07

Q.5	(a) As a Civil Engineer discuss your role in society	03
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- (b) Explain physical and chemical properties of cement **04**
- (c) Describes characteristics of contour in brief **07**

OR

- Q.5**
- (a) Give application of Remote sensing **03**
 - (b) Give Dimension of Modular Brick and discuss characteristics of first class brick **04**
 - (c) Explain concept and functions of smart cities . **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I &II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110004****Date: 11/06/2019****Subject Name: Basic Civil Engineering****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Discuss Role of Civil Engineer in society.	03
	(b) Explain traffic signals in brief.	04
	(c) Enlist various types of civil engineering materials. Discuss types, properties and uses of cement.	07
Q.2	(a) What is FSI? Discuss in brief.	03
	(b) Write a short note on low cost housing.	04
	(c) Enlist and explain principles of planning.	07
	OR	
	(c) Discuss objects and advantages of zoning.	07
Q.3	(a) What is building services? Explain in brief.	03
	(b) Draw any four symbols used in electrical layout.	04
	(c) Enlist and explain various types of loads acting on building.	07
	OR	
Q.3	(a) What is nominal dimensions for door and window?	03
	(b) What are different types of brick bonds? Explain any one.	04
	(c) Explain with neat sketch various components of building & its functions.	07
Q.4	(a) Enlist various instruments used in linear measurement.	03
	(b) What is whole circle bearing and quadrantal bearing?	04
	(c) Explain methods of levelling.	07
	OR	
Q.4	(a) Define: Contour, Meridian, Bearing.	03
	(b) Draw a neat sketch of a theodolite showing its various parts.	04
	(c) Explain chaining on plane ground.	07
Q.5	(a) Explain smart city and its features.	03
	(b) Compare BRTS and Metro.	04
	(c) Discuss Heritage structures & its conservations.	07
	OR	
Q.5	(a) Explain solid waste management system in brief.	03
	(b) Write a short note on rain water harvesting systems.	04
	(c) Describe mass transportation systems.	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110004****Date: 08/01/2020****Subject Name: Basic Civil Engineering****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Discuss the impact of infrastructural development on economy of country.	03
	(b) Enlist various types of cement and discuss the properties of any two.	04
	(c) Define Remote Sensing. Give the applications of remote sensing.	07
Q.2	(a) Define the following terms: a) Datum b) Change Point c) Contour Interval	03
	(b) Explain site plan and key plan.	04
	(c) Enlist various building services and objectives of building services.	07
	OR	
	(c) Draw a detailed sketch of a cross section of a wall showing the components of the building.	07
Q.3	(a) Give necessity of town planning.	03
	(b) Differentiate between plane survey and geodetic survey.	04
	(c) Enumerate various principles of planning and explain any two in detail.	07
	OR	
Q.3	(a) Draw traffic signals for (i) speed limit (ii) No parking	03
	(b) What are the benefits of Green building	04
	(c) Briefly outline causes characteristics and effects of slum.	07
Q.4	(a) Enlist Instruments used in chain surveying	03
	(b) Write short note on FSI	04
	(c) Describe water borne diseases with control measures.	07
	OR	
Q.4	(a) Give difference between Surveyors and prismatic compass.	03
	(b) Give full form of: a) DPC b) MH c) FSL d) BBCC	04
	(c) Explain various aids and devices used to control and regulate traffic.	07
Q.5	(a) Differentiate between W.C.B and Q.B system.	03
	(b) Explain with sketch the use of Line ranger.	04
	(c) Write short note on BRTS	07
	OR	
Q.5	(a) Describe with sketches characteristics of contours.	03
	(b) Write short note on origin of towns.	04
	(c) Given below are the bearings of lines of a closed traverse ABCD. Calculate the interior angles.	07

Line	F.B
AB	N45°E
BC	N75°E
CD	S35°W
DA	N65°W

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110004****Date:25/03/2022****Subject Name:Basic Civil Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- | | Marks |
|--|--------------|
| Q.1 (a) Write basic unit of measurements of length, area and volume in British and Metric system. | 03 |
| (b) What are the properties of good stone. | 04 |
| (c) What is intersection? Explain two broad groups of intersections with neat sketches. | 07 |
| Q.2 (a) Describe various principles of zoning. | 03 |
| (b) Draw and explain with neat sketch of an exogenous tree and show its component parts. | 04 |
| (c) Explain chemical characteristics of waste water. | 07 |
| OR | |
| (c) What are the elementary principles of building planning? Describe any two. | 07 |
| Q.3 (a) Discuss importance of land use planning. | 03 |
| (b) Write the general features of earthquake resistant building. Explain any one. | 04 |
| (c) The following consecutive readings were taken with a dumpy level along a chain line at a common interval of 5m. The position of the instrument was changed after 3 rd , 7 th and 10 th readings. Draw out the form of a level book and enter the readings properly. Assume RL of 1 st point as 50.0 meter. Calculate RL of all the points and apply usual checks. Use HI method. The readings are 0.350, 0.580, 0.645, 1.850, 1.935, 2.450, 1.750, 0.360, 0.720, 1.200, 0.920, 0.630. | 07 |
| OR | |
| Q.3 (a) Discuss functional requirements of residential building. | 03 |
| (b) Draw any four symbols used in electrical layout. | 04 |
| (c) Discuss characteristics of contours with suitable sketches. | 07 |
| Q.4 (a) Write types of offsets. | 03 |
| (b) What are different types of brick bonds? Explain any one. | 04 |
| (c) Explain various principles of river front development. | 07 |
| OR | |
| Q.4 (a) Functions and application of GIS. | 03 |
| (b) What do you mean by green building? Write its objectives. | 04 |
| (c) The following are the F.B & B.B of lines of a closed traverse. Calculate the interior angles of the traverse. | 07 |

LINE	FB	BB
AB	150°15'	330°15'
BC	20°30'	200°30'
CD	295°45'	115°45'
DE	218°	38°
EA	120°30'	300°30'

- Q.5 (a)** Write importance and objectives of providing house drainage. **03**
- (b)** What are the main reasons for the formation of slums? **04**
- (c)** Explain classification of buildings based on structure. **07**
- OR**
- Q.5 (a)** Define smart city. Which are the factors required to become a smart city? **03**
- (b)** Write various methods of disposal of solid waste. Explain land filling. **04**
- (c)** Write short note on BRTS. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022****Subject Code:3110004****Date:08-08-2022****Subject Name:Basic Civil Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- Q.1** (a) What is a role of civil engineer in the society? **03**
 (b) Explain the important pillars of smart city. **04**
 (c) The following bearings were taken on a closed traverse PQRS find out internal angles and correct them if necessary **07**

Line	FB	BB
PQ	40°00'	220°00'
QR	80°25'	260°25'
RS	145°50'	325°50'
SP	270°50'	90°50'

- Q.2** (a) Why the town planning is essential and necessary? **03**
 (b) Explain the primary divisions of surveying. **04**
 (c) Explain the classification of buildings in detail. **07**
OR
 (c) Explain the classification of surveying in detail. **07**

- Q.3** (a) Explain the types of meridians. **03**
 (b) What points are to be kept in mind while selecting the site for a residential building? **04**
 (c) Explain in detail features of smart city **07**

OR

- Q.3** (a) Enlist and explain types of bearing with neat sketch. **03**
 (b) Discuss the types of traffic signs and draw traffic sign for speed limit, no-parking and cross road. **04**
 (c) Explain the principles of building planning. **07**

- Q.4** (a) Write the advantages and disadvantages of riverfront development. **03**
 (b) Write a short note on growth of towns. **04**
 (c) Explain the impact of industrial development on the economy of the country. **07**

OR

- Q.4** (a) What is solid waste management? Discuss in brief the classification of solid waste. **03**
 (b) Discuss the concept of smart city. **04**
 (c) Enlist types of bonds in masonry. Draw a neat sketch of english bond and explain it **07**

- Q.5** (a) Discuss the main reasons for the slum formation. **03**

- (b) Discuss in brief the scope of civil engineering according to field of work. **04**
(c) Explain with neat sketch key plan, site plan, and layout plan. **07**

OR

- Q.5** (a) What is the concept of BRTS? Discuss its advantages and disadvantages. **03**
(b) Define zoning. Write objectives and principles of zoning. **04**
(c) What are the properties of good bricks? Discuss the classification of bricks. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022****Subject Code:3110004****Date:09-03-2023****Subject Name:Basic Civil Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) Discuss Branches of Civil Engineering. **03**
(b) Enlist principles of planning and describe any one in detail. **04**
(c) Enlist different types of foundation and explain Open footing. **07**

- Q.2** (a) Give necessity of town planning. **03**
(b) Differentiate between plane survey and geodetic survey. **04**
(c) Enlist various building services and objectives of building services **07**

OR

- (c) Draw a detailed sketch of a cross section of a wall showing the components of the building. **07**

- Q.3** (a) Enlist various instruments used in linear measurement. **03**
(b) What is whole circle bearing and quadrantal bearing? **04**
(c) The following consecutive readings were taken with a dumpy level and 4 m leveling staff on continuous sloping ground at 30 m intervals. 0.680, 1.455, 1.855, 2.330, 2.885, 3.380, 1.055, 1.860, 2.265, 3.540, 0.835, 0.945, 1.530 and 2.250. The RL of starting point was 80.750 m. Rule out page of a level book and enters the above readings. Find RL of other points by Rise and Fall method and apply necessary checks. Also determine gradient of a line joining first and last point. **07**

OR

- Q.3** (a) Give difference between Surveyors and prismatic compass. **03**
(b) Draw sketch of Dumpy level and show all parts **04**
(c) Explain various aids and devices used to control and regulate traffic. **07**

- Q.4** (a) Draw sketch of RCC beam. **03**
(b) Explain physical and chemical properties of cement. **04**
(c) Describes characteristics of contour in brief. **07**

OR

- Q.4** (a) Draw traffic signals for (i) speed limit (ii) No parking. **03**
(b) Write short note on FSI. **04**
(c) Enlist different bonds of Brick and Describe English bond in detail. **07**

- Q.5** (a) Explain smart city and its features. **03**
(b) Write a short note on rain water harvesting systems. **04**
(c) Describe mass transportation systems. **07**

OR

- Q.5** (a) Explain solid waste management system in brief. **03**
(b) Explain Energy efficient Building. **04**
(c) Write short note on BRTS. **07**

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023

Subject Code:3110004

Date:04-08-2023

Subject Name:Basic Civil Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Write the physical characteristics of waste water.	03
(b) Discuss the characteristics of good bricks.	04
(c) Describe the fundamental principle of surveying.	07
Q.2 (a) Draw the symbol for following material in section.	03
1. Earth	
2. Concrete	
3. Metal Section	
(b) Draw line diagram for 2 BHK residential building	04
(c) List the methods for natural growth and discuss ribbon development with their drawbacks in detail.	07
OR	
(c) Describe the remedial measures to prevent the slum.	07
Q.3 (a) Write the full name of following abbreviations.	03
1. R.C.C.	
2. G.L.	
3. F.S.I.	
(b) Describe principles of building planning related to elevation of building.	04
(c) List the types of cement and discuss any five in detail.	07
OR	
Q.3 (a) Do the unit conversion of followings.	03
1. 5 feet = _____ inches	
2. 10 feet square = _____ meter square	
3. 2000 centimeter cube = _____ liter	
(b) State difference between intersection at grade and grade separated intersection with neat sketch.	04
(c) Discuss the various construction techniques for low cost housing.	07
Q.4 (a) A 20m chain was found to be 10 cm long at the end of day's work after measuring 5000m. If the chain was correct before the commencement of the work, find the correct length of the lines.	03
(b) Differentiate between plane and geodetic surveying.	04

- (c) The following readings were taken with a level and 4m staff. Draw up a level book page and calculate the levels by height of instrument method. 0.578, 0.933, 1.768, 2.450, 3.005, 0.567, 1.181, 1.888, 3.679, 0.612, 0.705 and 1.810. The instrument was shifted after 5th and 9th reading. The R.L. of first station is 58.250m. **07**

OR

- Q.4** (a) Draw a plane scale of 1cm = 5m and show the 20m on it. **03**
 (b) Differentiate between prismatic compass and surveyor compass. **04**
 (c) The following bearings were observed with a compass for the closed traverse ABCDA. **07**

Line	FB	BB
AB	110°	290°
BC	190°	10°
CD	300°	121°
DA	20°	199°

- Where do you suspect the local attraction? Find the included angles.
Q.5 (a) Draw elevation and section of framed and panelled door. **03**
 (b) Draw a typical plan of drainage for small residential building. **04**
 (c) Differentiate between Flemish bond and English bond. **07**

OR

- Q.5** (a) List the general features of earthquake buildings and discuss dimensions of buildings in detail. **03**
 (b) Discuss the conservation strategies of heritage buildings. **04**
 (c) Differentiate between load bearing and framed structure. **07**

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023

Subject Code:3110004

Date:18-01-2024

Subject Name:Basic Civil Engineering

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Convert (i) 2 miles to kilometer (ii) 10 acres to square meter of area (iii) Express one cubic meter to cubic feet of volume	03
	(b) Explain batching of concrete and how it can be done	04
	(c) Draw a neat sketch of roof top rainwater harvesting system and explain how you will calculate the storage volume requirements.	07
Q.2	(a) State the permissible limits of the following parameters in drinking water for the following parameters as per IS Codes (i) BOD (ii) pH (iii) lead	03
	(b) Compare the properties of conventional bricks with stones in respect of fire resistance, water absorbing capacity and strength.	04
	(c) Discuss in detail how total station is a multipurpose instrument and explain its benefits with respect to speed and accuracy	07
	OR	
	(c) Discuss the importance of remote sensing and GIS as modern Surveying tools	07
Q.3	(a) Taking tread width for a flight of stairs to be 30 cm and individual risers to be 15 cm calculate the number of risers and treads for the vertical separation between the floors as 3.3 meters. State the number of riser after which landing will be provided.	03
	(b) Explain Floor Space Index and its importance in building planning.	04
	(c) Draw a proportionate line plan of a low-cost housing building having living room, bedroom, kitchen, washing platform and water closet. Mention sizes.	07
	OR	
Q.3	(a) Discuss how the organic waste generated from kitchens for a small town should be managed and disposed off.	03
	(b) Explain how proper orientation of a building can make it more energy efficient for the building located in tropical climate.	04
	(c) Draw cross section through door for a single wall for a residential building from foundation bottom to parapet top showing spread footing foundation for 30 cm thick wall. Label the components and provide dimensions.	07
Q.4	(a) Give the nominal dimension of a door in a residential building, chair for a single person and a single bed cot.	03
	(b) Discuss how you will decide the plinth level of a building.	04

- (c) The following consecutive readings were taken with a level and a 4meter levelling staff on a continuously sloping ground: 0.750, 1.535, 1.955, 2.430, 2.985, 3.480, 1.55 and 1.960. The reduced level of the first point was 100.750. The instrument was shifted after taking the sixth and the eighth staff reading. Calculate reduced level of all stations using the rise and fall method. **07**

OR

- Q.4** (a) Convert the following whole circle bearings into quadrantal bearings: (i) 30° (ii) 120° (iii) 340° **03**
- (b) Explain how water cement ratio affects the strength and workability of concrete. **04**
- (c) Bearings of a four-sided closed traverse were observed as per the table below: **07**

Table for Q4(c)		
Line	F.B.	B. B.
AB	$120^\circ 00'$	$300^\circ 00'$
BC	$40^\circ 00'$	$220^\circ 00'$
CD	$300^\circ 15'$	$120^\circ 15'$
DA	$200^\circ 45'$	$20^\circ 45'$

Check for local attractions if any and calculate the internal angles.

- Q.5** (a) Enlist three traditional building materials that were used in construction of historical heritage buildings of India. **03**
- (b) Explain the concept and advantage of locally available materials in construction of green buildings **04**
- (c) Discuss the structural measures taken which will make buildings less vulnerable to damage due to earthquakes. **07**

OR

- Q.5** (a) Compare the properties of fiber reinforced polymers and plastic and hence give the application and uses of FRP. **03**
- (b) What is the concept of BRTS? Discuss its advantages. **04**
- (c) Discuss the existing gaps and problems in conservation of heritage buildings in India. **07**

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110004

Date:29-06-2024

Subject Name:Basic Civil Engineering

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Explain role of Civil engineers.	03
	(b) Discuss various grades of Concrete.	04
	(c) What are the key components of GIS? Describe them briefly.	07
Q.2	(a) Define the following terms: a) Mean Sea Level b) Level Surface c) Bench Mark	03
	(b) Explain site plan and key plan.	04
	(c) Discuss various pipes used in the house drainage plumbing system.	07
	OR	
	(c) Draw a detailed sketch of a cross section of a wall showing the components of the building.	07
Q.3	(a) Discuss aims and objectives of Town planning.	03
	(b) Differentiate between plane survey and geodetic survey.	04
	(c) Enumerate various principles of planning and explain any two in detail.	07
	OR	
Q.3	(a) Draw traffic signals for (i) No Parking (ii) Over taking prohibited.	03
	(b) What are the benefits of Energy efficient building?	04
	(c) Describe the various principles of Zoning.	07
Q.4	(a) Differentiate between W.C.B and Q.B system.	03
	(b) Write short note on FAR.	04
	(c) Describe water borne diseases with control measures.	07
	OR	
Q.4	(a) Enlist Instruments used in chain surveying	03
	(b) Give full form of: a) MH b) FSL c) BBCC d) RCC	04
	(c) Describe pillars of smart city.	07
Q.5	(a) Discuss different types of Bearings.	03
	(b) Explain with sketch the use of Line ranger.	04
	(c) Explain various aids and devices used to control and regulate traffic.	07
	OR	
Q.5	(a) Describe with sketches characteristics of contours.	03
	(b) What are the temporary adjustments of Level? How are they done?	04
	(c) The following are the fore bearings of a closed traverse. Calculate back bearings and interior angles of the traverse.	07

Line	F. B
AB	N 45°30' E
BC	S 60°00' E
CD	S 10°30' W
DA	N 75°45' W

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:3110004****Date:08-01-2025****Subject Name:Basic Civil Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Enlist different branches of civil engineering and explain any two.	03
	(b) Discuss properties of cement.	04
	(c) What are the key components of GIS? Describe them briefly.	07
Q.2	(a) Convert the following WCB in to RB: 1) 55°30' 2) 279°00' 3) 190°00'	03
	(b) Explain primary division of surveying.	04
	(c) Discuss classification of buildings based on structure.	07
OR		
	(c) Write short note on HVAC.	07
Q.3	(a) Discuss aims and objectives of town planning.	03
	(b) Describe pillars of smart city.	04
	(c) Describe the important remedial measures to prevent slum formation.	07
OR		
Q.3	(a) Discuss traffic characteristics.	03
	(b) Explain site plan and key plan.	04
	(c) Enumerate various principles of planning and explain any two in detail.	07
Q.4	(a) What are the various types of loads acting on building.	03
	(b) Discuss importance of zoning.	04
	(c) Discuss rooftop rain water harvesting system with neat sketch.	07
OR		
Q.4	(a) Enlist the various instruments used in chaining.	03
	(b) Differentiate between prismatic compass and surveyor compass.	04
	(c) Discuss the general features of earthquake resistant buildings.	07
Q.5	(a) What are the different types of meridians?	03
	(b) Explain the working principle of Line ranger.	04
	(c) The following are bearings of a closed traverse ABCDA. Find out interior angles and correct them if necessary.	07

LINE	FB	B.B
AB	45°00'	225°00'
BC	123°30'	303°30'
CD	181°00'	1°00'
DA	289°00'	109°00'

OR

- Q.5** (a) What are the uses of contour map? **03**
 (b) Describe various types of benchmarks. **04**
 (c) The following observations were taken with dumpy level and four-meter levelling staff. The instrument was shifted after 4th and 7th readings. The first reading was taken on a B.M whose RL is 15.575m, prepare a page of level book and calculate RLs of all the points. The observation was taken at every 30m interval. Also find the gradient between first and last point. **07**
 The observations are: 0.565, 1.250, 1.675, 3.695, 0.125, 2.345, 3.245, 0.500, 1.785, 2.535.

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:3110004

Date:30-06-2025

Subject Name:Basic Civil Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	MARKS
Q.1 (a) What is FSI? Discuss in brief.	03
(b) Write a short note on low cost housing.	04
(c) Enlist and explain principles of planning.	07
Q.2 (a) Describe primary divisions of surveying.	03
(b) Enlist various types of cement and discuss the properties of any two.	04
(c) Explain with neat sketch various components of building & its functions	07
OR	
(c) Enlist different bonds of Brick and Describe English bond in detail	07
Q.3 (a) Explain Application of Total station.	03
(b) Explain traffic signals in brief.	04
(c) Enlist various building services and objectives of building services.	07
OR	
Q.3 (a) As a Civil Engineer discuss your role in society	03
(b) Draw any four symbols used in electrical layout.	04
(c) Give classification of solid waste and describe why its segregation in wet and dry is necessary?	07
Q.4 (a) What is nominal dimensions for door and window?	03
(b) Draw sketch of Dumpy level and show all parts.	04
(c) Write short note on BRTS.	07
OR	
Q.4 (a) Enlist Instruments used in chain surveying.	03
(b) Write a short note on rain water harvesting systems	04
(c) Describes characteristics of contour in detail.	07
Q.5 (a) Give application of Remote sensing	03
(b) Compare BRTS and Metro.	04
(c) Define : (1) offsetting (2) Base line (3) Magnetic Meridian (4) True Bearing (5) Reduced Bearing (6) Bench Mark (7) Mean Sea Level	07

OR

- Q.5** (a) Give Dimension of Modular Brick and discuss characteristics of first class brick **03**
- (b) Give full form of: a) DPC b) MH c) FSL d) BBCC **04**
- (c) Discuss Heritage structures & its conservations. **07**

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: 3110004

Date: 09-01-2026

Subject Name: Basic Civil Engineering

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- | | Marks |
|--|--------------|
| Q.1 (a) Define Civil Engineering and state its role in infrastructural development | 03 |
| (b) Enlist different types of cement. Explain properties of OPC and PPC. | 04 |
| (c) Describe properties and uses of any three: (i) Timber, (ii) Plastic, (iii) Fly ash, (iv) Bitumen. | 07 |
| Q.2 (a) Define: (i) Site plan (ii) Key plan (iii) Elevation. | 03 |
| (b) State and explain any two principles of planning a residential building. | 04 |
| (c) Discuss the necessity of town planning and advantages of zoning. | 07 |
| OR | |
| (c) Explain causes, characteristics and effects of slums. Suggest preventive measures. | 07 |
| Q.3 (a) Enlist types of building loads with examples. | 03 |
| (b) Differentiate between English bond and Flemish bond in brickwork | 04 |
| (c) Draw a neat labeled cross-section of a wall showing major building components | 07 |
| OR | |
| (a) Define building construction and list its main components. | 03 |
| (b) Explain the purpose of symbols in electrical and plumbing layouts. | 04 |
| (c) Enlist different types of foundation and explain Open footing. | 07 |
| Q.4 (a) Define: (i) Contour interval (ii) Benchmark (iii) Change point. | 03 |
| (b) Differentiate between WCB and QB system of bearings. | 04 |
| (c) The following bearings were taken of a closed traverse ABCD. Calculate the interior angles of the traverse. | 07 |

Line	F.B.	B.B
AB	45° 00'	225° 00'
BC	123° 30'	303° 30'
CD	181° 00'	1° 00'
DA	289° 00'	109° 00'

OR

- (a)** Define ranging in surveying. Why is it required? **03**
- (b)** Differentiate between fly leveling and differential leveling. **04**
- (c)** The following readings are taken on continuously falling ground with staff of 4 m. They are 0.400, 0.765, 1.270, 2.560, 3.220, 3.950, 0.390, 1.690, 3.500, 0.800, 1.920, 2.450, 3.980. Enter the readings in the page of level book and calculate the RLs of all the points if the first reading was taken on Benchmark of 100.00 m using Height of Instrument Method. **07**

- Q.5**
- (a)** Define Green building. List any three of its benefits. **03**
 - (b)** Write short note on features of Smart City. **04**
 - (c)** Explain Rainwater harvesting system with neat sketch. **07**

OR

- (a)** What is solid waste management? Give two methods of waste disposal. **03**
- (b)** Discuss the role of riverfront development in urban planning. **04**
- (c)** What is earthquake-resistant structures? Explain two principles used in their design. **07**
